



CARGO CONVOY CRM

Complete User Manual & Operations Guide



Cargo Convoy Inc

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Version 3.0 | July 2026 | Laravel 12 | PHP 8.4
305 Routes | 15 Controllers | 34 Models | 123 Views

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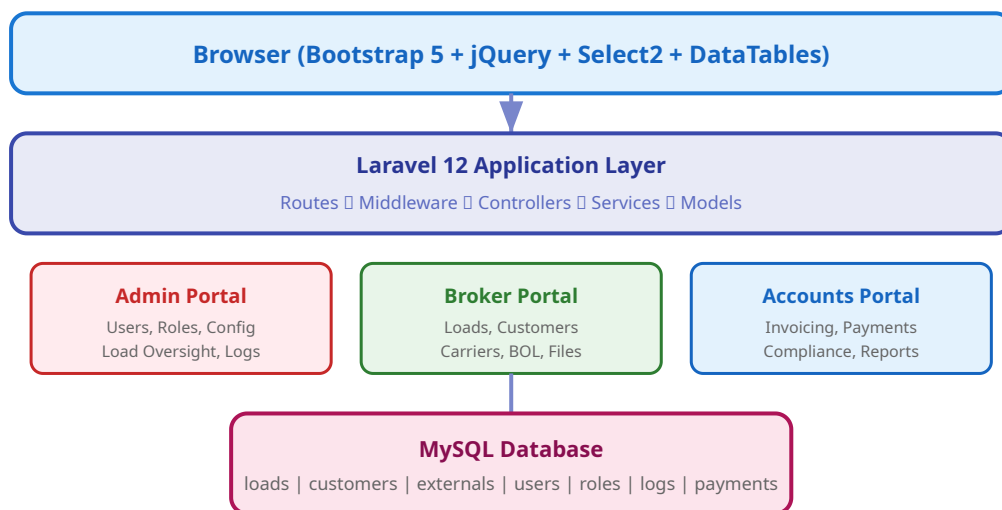
1. System Overview & Architecture

1.1 What is Cargo Convoy CRM?

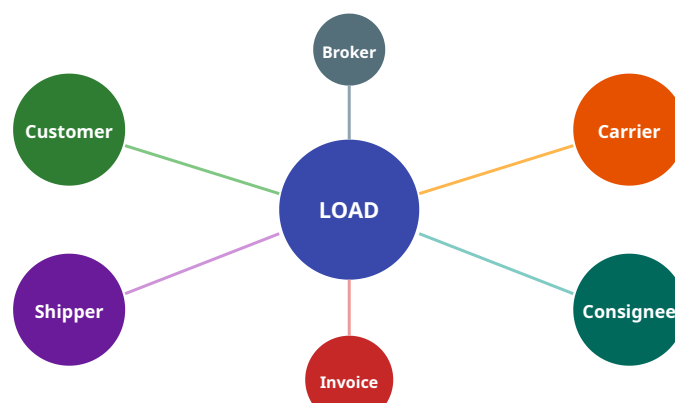
Cargo Convoy CRM is a comprehensive **freight brokerage management platform** built for Cargo Convoy Inc. It handles the entire lifecycle of freight operations — from customer onboarding and credit management, through load creation and carrier dispatch, to invoicing and payment collection.

Technology Stack: Laravel 12 (PHP 8.4) | MySQL 8 | Blade Templates | Bootstrap 5 | jQuery | Select2 | DataTables | Dompdf | FPDF | PhpSpreadsheet | ApexCharts

1.2 System Architecture

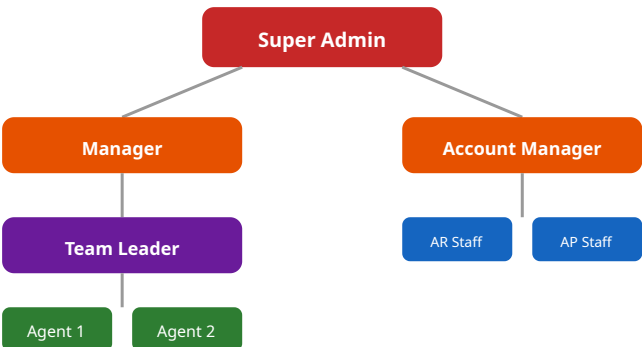


1.3 Module Interaction Map



2. User Roles & Permissions

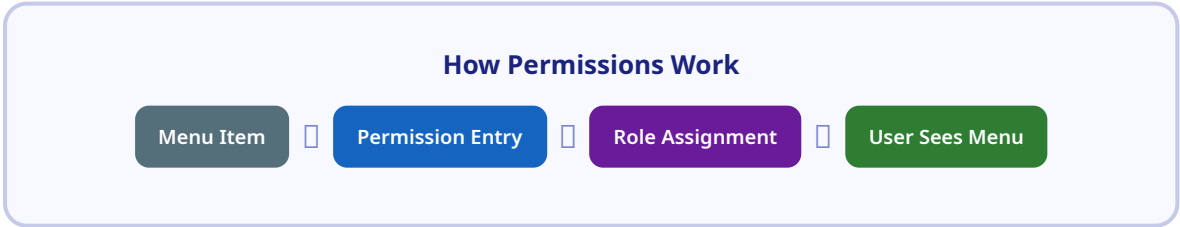
2.1 Organization Hierarchy



2.2 Role Access Matrix

Role Group	Role IDs	Portal Access	Data Visibility
Super Admin	1, 2, 3, 22	Admin + Broker + Account	All data system-wide
AR Accounting	7, 8, 9, 24	Account + Broker	All loads for invoicing
AP / Vendor	4, 5, 6	Account + Broker	Carrier payments
Compliance	10, 11, 12	Account + Broker	All carriers for review
Reporting	13-18	Account + Broker	Read-only reports
Manager	19	Broker only	Their TLs + agents data
Team Leader	20	Broker only	Their agents' data
Agent/Broker	21	Broker only	Own data only

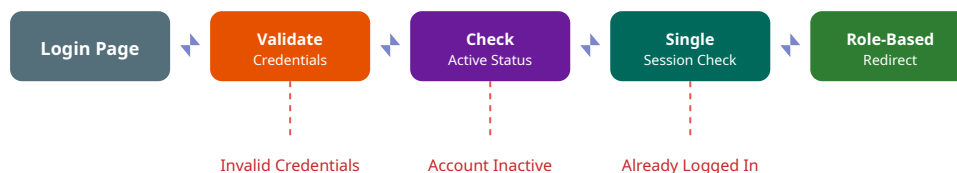
2.3 Permission System



Each menu item corresponds to a permission record. Roles are granted read/write access per permission. The sidebar menu dynamically shows only items the user's role has access to.

3. Authentication & Security

3.1 Login Flow



3.2 Security Features

Session Management

- Single active session per user
- Auto-logout on duplicate login
- Session timeout configurable
- LogoutOtherDevices enforced

Access Control

- Role-based URL restriction
- Menu visibility per permission
- Middleware validates every request
- Admin can reset any password

3.3 Activity Logging

Every significant action is recorded in the logs table with:

Who

- User name & ID
- User email
- Role

What

- Action description
- Old data (JSON)
- New data (JSON)

When/Where

- Timestamp
- IP address
- Full URL

4. Admin Panel

4.1 Admin Dashboard

Consolidated view of all loads across the system. Admins see data from all agents, offices, and teams with advanced filtering.

4.2 User Management Flow



4.3 Master Data Configuration

Equipment Types

- Dry Van
- Flatbed
- Reefer
- Step Deck
- Power Only

Shipment Types

- FTL (Full Truck)
- LTL (Less Than)
- TONU
- Partial
- Drayage

Organization

- Departments
- Offices
- Managers
- Team Leaders
- Countries/States

4.4 Admin Load Edit

Admins can edit any load in the system with full rate modification capability. Credit is automatically recalculated when rates change. Status transitions are validated.

5. Broker Module

5.1 Broker Workspace Overview

MENU

- Home
- Load Creation
- Customer
- Carrier
- Shipper
- Consignee

CARGO CONVOY INC - Broker Dashboard

All

Open

Delivered

Complete

Invoiced

Paid

Load#	Customer	Carrier	Rate	Status	Actions
33621	ABC Corp	Fast Carrier	\$2,500	Open	
33620	XYZ Inc	Quick Haul	\$1,800	Delivered	

5.2 Module Features

Load Management

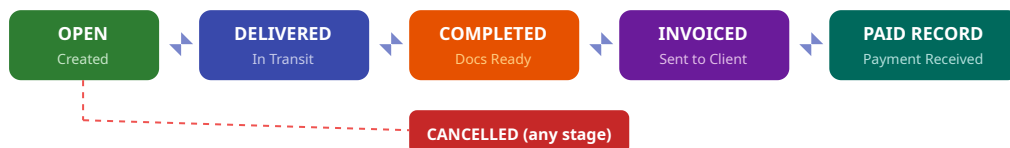
- Create new loads with multi-stop
- Edit rates, carrier, status
- Clone loads for repeat shipments
- Real-time credit validation
- Search & filter by all fields

Customer & Carrier

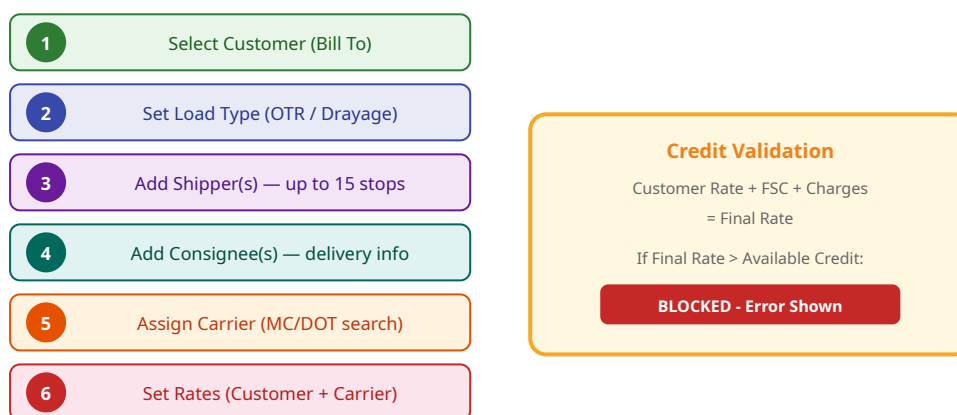
- Add/edit customers with approval flow
- Monitor credit limit usage
- Auto-suggest carriers by MC/DOT
- Upload remittance documents
- Cross-reference shipper/consignee

6. Load Management (Core Module)

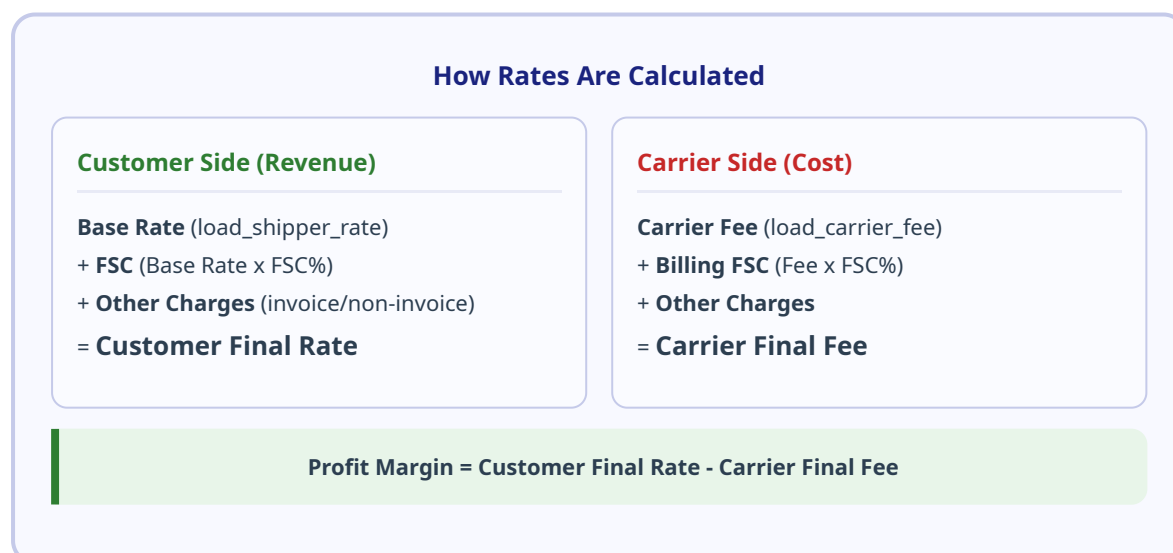
6.1 Load Lifecycle



6.2 Load Creation — Step by Step



6.3 Rate Calculation Formula



6.4 Load Clone Feature

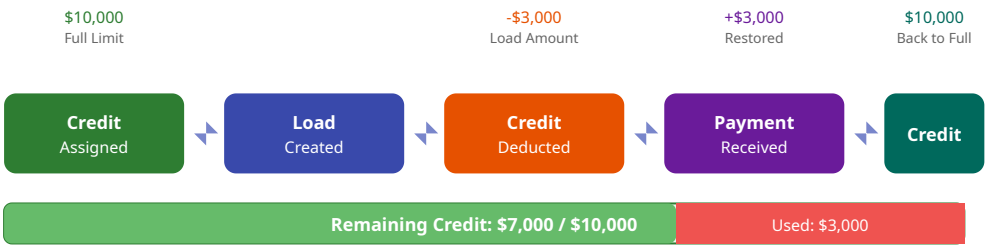
Quickly duplicate a load for repeat shipments. Copies all shipper, consignee, customer, and carrier data. Resets rates to 0, status to "Open", and assigns a new load number.

6.5 Load Tabs & Filtering

Tab	Filter	Description	Who Uses
All Loads	None	All loads paginated (50/page)	Broker, Admin
Open	load_status = 'Open'	Active awaiting dispatch	Broker
Delivered	load_status = 'Delivered'	Confirmed delivered	Broker, Accounts
Complete	load_status = 'Completed'	Ready for invoicing	Accounts
Invoiced	invoice_status = 'Paid'	Invoice sent to customer	Accounts
Paid	invoice_status = 'Paid Record'	Payment received	Accounts

7. Credit Limit System

7.1 Credit Lifecycle



7.2 Credit Fields Explained

Field	Type	Purpose	Who Sets It
adv_customer_credit_limit	float	Total assigned credit limit	Accounts team (via credit logs)
remaining_credit	float	Available balance (auto-calculated)	System only (never manual)
remaining_credit_amount	float	Synced copy for reporting	System only
invoice_credit_limit	float	Separate limit for invoice charges	Accounts team
credit_limit_log	JSON	Audit trail of all limit changes	Auto-recorded
remaining_credit_logs	JSON	Audit trail of balance changes	Auto-recorded

7.3 Server-Side Validation (Security)

Atomic Credit Check with Row Locking

Lock Customer Row



Check Balance



Deduct Amount



Save & Release

```
// CreditService.php — All credit operations use this pattern:
DB::transaction(function() {
    $customer = Customer::whereKey($id)->lockForUpdate()->first();
    $available = calculateAvailableCredit($customer);

    if ($loadAmount > $available) {
        throw new CreditExceededException("Available: $available, Requested: $loadAmount");
    }

    $customer->remaining_credit = $available - $loadAmount;
    $customer->remaining_credit_amount = $customer->remaining_credit;
    $customer->save();
});
```

Race Condition Protection: The lockForUpdate() prevents two simultaneous requests from reading the same balance. If two brokers create loads at the same time, one will wait for the other to finish — preventing double-spending.

8. Customer Management

8.1 Customer Approval Workflow



Only APPROVED customers can be selected when creating loads

8.2 Customer Information

Basic Details

- Customer Name
- MC/FF Type & Number
- Corporation Type
- Status (Approved/Not Approved)
- Hold/Unhold Status

Financial

- Credit Limit (Assigned)
- Remaining Credit (Computed)
- Payment Terms
- Factoring Company
- Invoice Through setting

Contact Info

Address

- Primary & Secondary Contact
- Email, Phone, Fax
- Billing Contact
- Billing Email & Phone

- Main Address (Street/City/State/ZIP)
- Billing Address (separate)
- Country & State dropdowns

9. Carrier Management

9.1 Carrier Verification Flow



9.2 Carrier Data & Auto-Suggest

When typing carrier name, MC#, or DOT# in the load form, the system provides **live AJAX suggestions**. Only MC-approved and unblocked carriers appear. Selecting auto-fills all fields.

Field	Validation
MC Number	Must be unique across all carriers
DOT Number	Must be unique across all carriers
Carrier Block	Blocked carriers cannot be assigned to loads
MC Check Status	Only "Approved" carriers show in suggestions

10. Shipper & Consignee

10.1 Multi-Stop Support

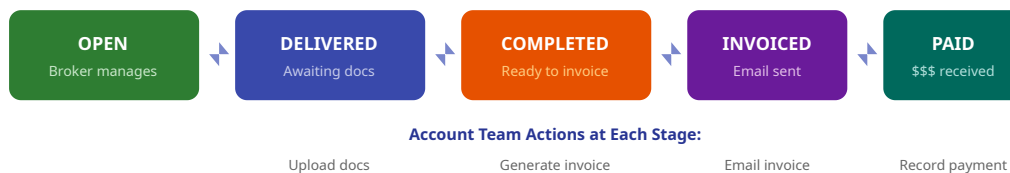


10.2 Cross-Reference Feature

When creating a Shipper, you can check "Also add as Consignee" (and vice versa). This saves time for locations that serve both roles.

11. Accounting & Invoicing

11.1 AR Accounting Pipeline



11.2 Invoice Email Process

- 1 Select Documents:** Choose which load documents to attach (Rate Con, POD, Invoice, DO)
- 2 Set Recipients:** Enter TO emails (comma-separated) + optional CC
- 3 Auto PDF Merge:** System automatically merges all selected documents into one PDF
- 4 Custom Subject:** Edit subject line or use default (Load # + Invoice # + Ref #)
- 5 Send:** Email sent via SMTP from ar@cargoconvoy.co with merged attachment

11.3 Carrier Payment Management

Payment Options

- **Quick Pay:** Mark for expedited payment
- **Payment Method:** ACH / Check / Zelle
- **Ready to Pay:** Flag when approved
- **Due Date:** Set carrier payment deadline
- **Mark as Paid:** Record completion

Carrier Documents

- Upload carrier invoices
- W9, Insurance, Authority Letter
- Bank verification documents
- Rate confirmation signed copies

11.4 AR Aging & Close

The aging report tracks days since invoice was sent. Account team can:

- View aging days per load
- Track excess/remaining amounts after partial payments
- Close aging records when payments are reconciled
- Filter by office, agent, and date range

11.5 All Load Status AR View

A comprehensive dashboard showing **all loads across all statuses** for the AR team with advanced filtering and bulk actions. Includes new aging close functionality.

12. Compliance Module

12.1 MC Verification Process



12.2 CPR Status Table

Status	Meaning	Action
Not Approved	Default — not yet reviewed	Compliance reviews carrier docs
Verified	Carrier approved for this load	Load can proceed
Not Verified	Carrier rejected	Needs carrier replacement

12.3 Carrier Block/Unblock

Compliance team can **block** carriers who fail checks. Blocked carriers are excluded from auto-suggest and cannot be assigned to new loads.

13. Reporting & Analytics

13.1 Report Types



13.2 Sales Dashboard

Interactive ApexCharts showing daily revenue trends. Filters: date range, office, agent. Displays: Customer Revenue (green), Carrier Cost (red), Margin (blue).

13.3 Export Options

All reports support **Excel export** via PhpSpreadsheet. Select any report → click "Excel" button → downloads .xlsx file with full data.

14. Email & PDF Merge

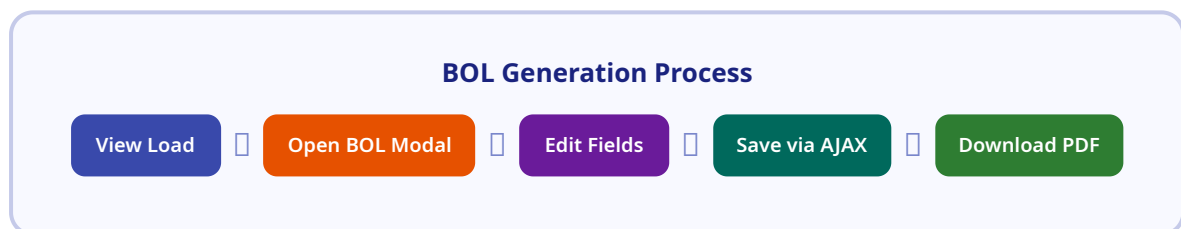
14.1 Email Flow



PDF Merge: Uses FPDF library to merge multiple PDFs + Dompdf to convert images to PDF. All selected docs become one single attachment.

15. Bill of Lading (BOL)

15.1 BOL Workflow

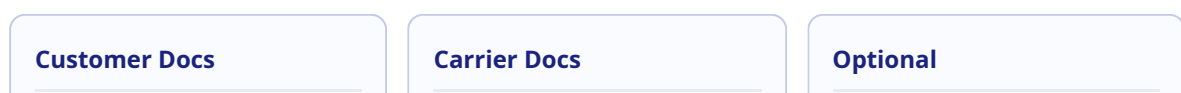


15.2 BOL Sections

Section	Content
Header	Company logo, name, address, phone, MC#
Load Info	Load#, BOL#, Ship Date, Delivery Date
Shipper/Consignee	Names and full addresses
Freight Table	Pieces, Description, Weight, Type, NMFC, Class
Notes & COD	Special instructions, COD amount, declared value
Signatures	Shipper, Carrier, Consignee signature lines with dates

16. Document Management

16.1 Document Types Per Load



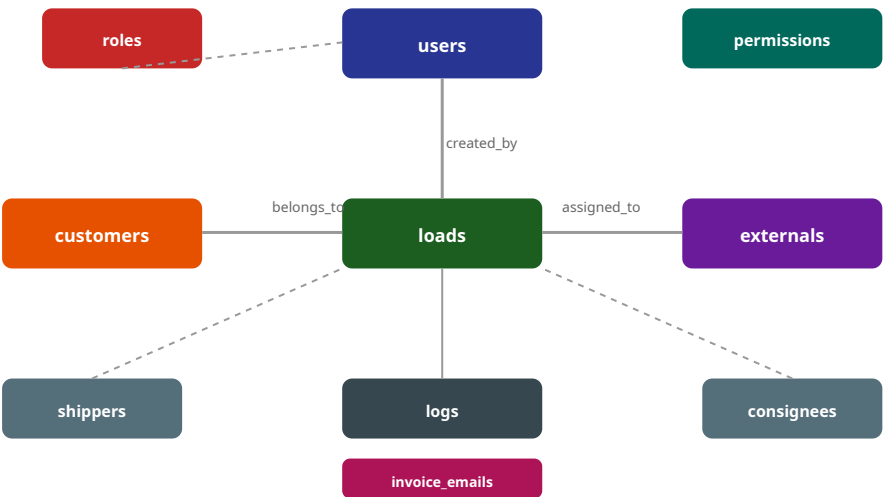
- Rate Confirmation
 - Shipper Rate Approval
 - Delivery Order (DO)
- Carrier Invoice
 - POD (Proof of Delivery)
 - W9 / Insurance
- Additional supporting docs
 - Multiple file upload
 - PDF + JPG + PNG accepted

16.2 File Storage Structure

```
public/uploads/  
  Load-files/{load_number}-{customer_id}/  Customer load documents  
  delivery-order/{load_id}/                  Private delivery orders  
  carrier/                                   Carrier verification docs  
  customers/{customer_name}/                 Customer files  
  remittances/                               Remittance uploads  
  mergedfiles/                               Auto-merged PDFs for email
```

17. Database Schema

17.1 Entity Relationship Diagram



17.2 Key Tables

Table	Records	Purpose
users	All system users	Authentication, roles, department assignment
loads	Freight loads	Core business entity — everything revolves around loads
customers	Customer companies	Bill-to entities with credit limits
externals	Carriers	Transport companies assigned to haul loads
shippers / consignees	Locations	Pickup and delivery locations
roles / permissions	Access control	Role-based menu and feature access
logs	Audit trail	Every create/update/delete logged with old/new data

18. API Endpoints Reference

18.1 Key Broker Endpoints

Method	Endpoint	Purpose
POST	/broker/load-create	Create new load (with credit check)
GET	/broker/load	Load listing with tabs
POST	/broker/load/update/{id}	Update load (rate change + credit delta)
GET	/broker/clone/load/{id}	Clone a load
GET	/broker/load/{id}/bol/pdf	Download BOL as PDF
POST	/broker/customer/create	Create customer
POST	/broker/create/carrier	Create carrier

18.2 Key Account Endpoints

Method	Endpoint	Purpose
GET	/account/accounting	AR accounting dashboard
POST	/account/send-mail	Send invoice email (with PDF merge)
POST	/account/update-invoice-status/{id}	Mark as invoiced
POST	/account/update-ar-aging-close	Close aging record
GET	/account/reporting	Reports & analytics
GET	/account/credit	Credit management view
GET	/account/compliance	MC/CPR compliance
PUT	/account/update-customer/{id}	Update customer (credit recalculated)

